INCH-POUND
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(See 6.8)

MILITARY SPECIFICATION

CLEANING SYSTEM, ULTRASONIC

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ultrasonic cleaning systems. The systems are electrically powered with equipment and accessories as specified herein.

1.2 Classification. The cleaning system covered by this specification shall be of the size and capacity shown in tables I and II, as specified (see 6.2.1).

TABLE I. Classification.

System size	Tank capacity, gallons, nominal	Root mean square (rms) output continuous minimum, average, watts
1	1.5	500
2	5	1000
3	12	2000
4	25	3000
Special	As specified	As specified

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE II. Cleaning and rinsing tanks dimensions and capacity.

Internal working dimensions[1] inches	Systems size[2]			
	1	2	3	4
Length	9	14	18	25
Width	66	9	12	18
Depth (liquid)	5	9	10	12

[1] Dimensions shall be within plus or minus 1/4 inch.
[2] Special size, dimensions to be as specified.

1.3 Requirements for cleaning systems of a size and capacity not shown in tables I and II shall be designated "special". Deviations or modifications shall be fully described, as specified (see 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks term a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-C-2212 - Controllers, Electric Motor A.C. or D.C. and Associated Switching Devices.
MIL-E-5007 - Engines, Aircraft, Turbojet and Turbofan, General Specification for.
MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.
MIL-D-16791 - Detergents, General Purpose (Liquid, Nonionic).
MIL-B-17931 - Bearings, Ball, Annular, for Quiet Operation.
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.

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MILITARY (Continued)

MIL-T-20168 - Tubes, Brass (Red Brass), Seamless.
MIL-H-22577 - Heating Elements, Electrical: Cartridge, Strip and Tubular Type.
M1L-T-24107 - Tube, Copper, (Seamless).

STANDARDS

MILITARY

MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-470 - Maintainability Program for Systems and Equipment.
MIL-STD-740-1 - Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.
NIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-2073-1- DoD Materiel Procedures for Development and Application of Packaging Requirements.

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.
MIL-HDBK-472 - Maintainability Prediction.

(Copies of specifications, standards and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AIRCRAFT POROUS MEDIA, INC.
Field Service Report No. 10C - Evaluating Effectiveness of Filter Element Cleaning Methods.

(Application for copies should be addressed to the Aircraft Porous Media, Inc., 32 Sea Cliff Ave., Glen Cove, New York, NY 11542.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Construction. The cleaning system shall remove contaminants from service dirtied air filters, electrostatic precipitator cells, lube-oil and hydraulic system filters, valves and other components within its capacity. The system shall consist of an ultrasonic generator, transducers, cleaning tank, rinse tank, recirculating system, cabinets and controls. The recirculating system shall incorporate a filter, heater, and pump. Means shall be provided to control and adjust cleaning solution temperature. A safety thermostat shall also be provided (see 3.2.5). The system shall be for use with aqueous detergent solutions and water rinsing. System components requiring maintenance or replacement shall be readily accessible from the top or front of the unit or cabinet. The equipment shall be constructed of parts that are new, without defects, and free of repair. The equipment shall be complete so that when connected to the specified utilities, it can be used for any operation for which it is being acquired.

3.2.1 Cleaning. The system shall meet the cleaning tests as specified (see 4.6.4, 4.6.5, and 4.6.6).

3.2.2 Inclination. The system shall be so constructed that inclination of the equipment of 30 degrees in any direction shall not affect operation or cause spillage of the cleaning solution from the tank. Liquid fill limit shall be indicated by a permanent mark or controlled by an overflow outlet.

3.2.3 Submarine hatch access. When specified for submarine installation (see 6.2.1), the components of the system shall be of such size or construction

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that they can be passed through a circular submarine hatch. Size restriction shall be as specified (see 6.2.1). If the component is too large for this passage, it shall be possible to take it apart, pass the parts through the hatch, and reassemble the system in the submarine. If necessary, this assembly shall be under the supervision of a contractor's representative.

3.2.4 Noise limitations (airborne). Unless otherwise specified (see 6.2.1), airborne noise requirements shall be in accordance with the measurement and acceptance criteria for grade designation B as specified in MIL-STD-740-1. Airborne noise shall be limited to 75 decibels.

3.2.4.1 Noise limitations (structureborne). Unless otherwise specified (see 6.2.1), the equipment shall be provided with mounting holes and hardware compatible with sound isolation mounts (Navy type 7E450). Structureborne noise requirements shall be in accordance with MIL-STD-740-2.

3.2.5 Safety devices. The system shall be fitted with safety devices, covers or doors complete with electrical interlocks for all parts that present safety hazards as specified in MIL-E-16400. Interlocks and guards shall be constructed to provide access. The system shall be grounded (excluding electrical power source).

3.2.6 Bearings. Bearings shall be of a sealed type in accordance with MIL-B-17931.

3.2.7 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.2.8 Maintainability. The equipment shall be constructed to permit maintenance personnel to easily and effectively service the equipment using a minimal number of tools.

3.3 Castings and forgings. Castings and forgings shall be free from defects, scale and mismatching. Processes such as welding, peening, plugging, or filling with cold solders or metallic pastes shall not be used on castings or forgings for reclaiming any parts of the equipment.

3.3.1 Welding, brazing or soldering. Welding, brazing or soldering shall be employed only where those operations are included in fabrication of the original construction. These operations shall not be employed as repair measures for defective parts.

3.3.2 Soldering. Solder connections shall show evidence of good bonding in metal-to-metal contact. Cold solder joints, incomplete joining of solder and metal, excess or insufficient solder or damaged insulation shall be considered sufficient reason for rejection of the equipment.

3.3.3 Cleaning. Any loose, spattered solder, flux, metal chips, insulation scrap or other foreign material shall be removed from the equipment.

3.3.4 Flux and cleaning agents. Flux for soldering shall be rosin or rosin and alcohol. No acid, salts or acid core solder shall be used in preparation for soldering of electrical connections.

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3.3.5 Fastening devices. Fastening devices shall be in accordance with MIL-STD-1754 and shall be installed with means for preventing change of tightness. Such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed. Fastening devices shall be tightened to torque limits as established by the manufacturer's standard for tightening to preclude loosening by normal operation or vibration.

3.3.6 Surfaces. Surfaces shall be cleaned and free from dirt, fins, sprues, scale, flux or other harmful or extraneous materials. External surfaces shall be smooth and the edges shall be either rounded or beveled.

3.3.7 Painting. Unless otherwise specified (see 6.2.1), parts other than corrosion resistant steel shall be painted in accordance with the contractor's standard commercial practice.

3.3.8 Threads. Threaded parts shall be in the inch system and conform to MIL-STD-1251. Tapered pipe threads shall not be permitted for shipboard equipment.

3.3.9 Shock. The equipment shall meet the requirements of grade B, class I equipment for electrical enclosures, and class II for cleaner and rinse consoles, medium weight, as specified in MIL-S-901.

3.3.10 Vibration. The equipment shall meet the requirements for type I in accordance with MIL-STD-167-1.

3.3.11 Mercury. The equipment shall contain neither mercury or mercury components, nor be exposed to free mercury during manufacture.

3.3.12 Caution or warning plates. A caution or warning plate shall be furnished on the equipment in a visible location. The caution or warning plate shall include the following: "The use of "Freon" or other solvent in this equipment is strictly prohibited."

3.4 Components. The equipment shall consist of, but shall not be limited to, the following principal components, attachments and accessories necessary to meet the operational and performance requirements specified herein.

3.4.1 Ultrasonic generator. The ultrasonic generator shall conform to the requirements of MIL-E-917 and MIL-E-16400. The generator shall provide sufficient energy in conjunction with the transducers to meet cleaning requirements of this specification and shall have an average continuous output as specified in table I at a frequency between 19 and 21 kilohertz. The frequency shall be manually tunable to a fixed frequency for maximum performance. Solid state electronic parts and other components shall be incorporated in capacities safely below their rated maximum output for optimum reliability. Physical and electronic protection shall be included for insurance against catastrophic failure through exposure to liquid or dry tank operation.

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3.4.2 Transducers. The transducers shall be zero spaced, nickel laminate, magnetostrictive stacks, geometrically arranged on the tank bottom to ensure uniform energy distribution in the cleaning tank. The transducers' stacks shall be silverbrazed to a 316L stainless steel tank bottom. The transducer coils shall be made field replaceable.

3.4.3 Cleaning tank. The cleaning tank shall hold objects being cleaned in the system. Unless otherwise specified (see 6.2.1), the tank shall be of the minimum working dimensions for the size specified in table II. The tank shall be made from 316L stainless steel. The inside of the tank shall have a no. 63 finish. The tank shall be arranged so that long objects may be cleaned by being immersed one half at a time. The tank shall have the necessary fittings for attaching to the recirculation system. The tank shall be fitted with a drain connection and a drain valve. The tank shall be marked showing the proper fluid level in order to avoid spillage when inclined, as specified in 3.2.2.

3.4.4 Rinsing tank. A rinsing tank shall be provided for rinsing objects after cleaning. The tank shall be made of 316L stainless steel. Unless otherwise specified (see 6.2.1), the tank shall have the minimum working dimensions specified in table II for the size specified. The inside of the tank shall have a no. 63 finish. The rinsing tank shall be arranged so that long objects may be rinsed by being immersed one half at a time. The tank shall be fitted with a drain connection and drain valve. The tank shall have provisions for handling the overflow during rinsing and the overflow shall be brought out separately from the drain system.

3.4.5 Recirculating system. When specified (see 6.2.1), a recirculation system shall be provided to pump, filter, and heat the solution in the ultrasonic cleaning tank. The system shall consist of a pump, filter, heater and connecting piping. The system shall have a minimum flow rate of 50 percent of the tank volume per minute. Means shall be provided to shut off the heater and pump when the system filter is clogged.

3.4.5.1 Pump. A stainless steel pump suitable for use with all commonly used aqueous cleaning solutions shall be provided. The pump shall be directly driven by an electric motor rated for continuous duty without exceeding the contractor's allowable temperature rise. The motor shall be protected from damage by fluids. The motor shall operate properly after being subjected to the shock and vibration tests (see 4.6.14 and 4.6.15).

3.4.5.2 Filter. A filter or filters (throw away element) and strainer or strainers shall be provided to maintain a useful cleaning solution life span. The strainer/filter capacity and retention capability shall be governed by the intended use of the ultrasonic cleaning system and shall be as specified (see 6.2.1). The filter shall be fitted with a device to indicate the need for filter cartridge replacement and to remove power to the heater and recirculating pump when the filter is clogged (see 3.4.7). The filter cartridge shall be readily accessible for replacement without emptying the cleaning tank.

3.4.5.3 Heater. An electric heater for heating the cleaning solution in accordance with type III of MIL-H-22577 shall be provided in the recirculating system. The heater tank and heater element sheaths shall be of nonmagnetic

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stainless steel. The heater shall raise the temperature of the full cleaning tank from 60 to 180 degrees Fahrenheit (°F) within 120 minutes.

3.4.5.3.1 Safety thermostat. A safety thermostat shall be furnished in the heater tank (well mounted) and shall operate at 185 ± 5°F (see 3.2).

3.4.5.4 Operating thermostat. An operating thermostat and a stainless steel sensing bulb shall be mounted in the recirculating solution piping. The thermostat adjusting knob shall be readily accessible to the-operator and shall operate from at least 50 to not less than 200°F.

3.4.5.4.1 Temperature controller. When specified (see 6.2.1), a solid state time-proportioning temperature controller shall be provided to control the temperature of the cleaning solution. The controller shall be in accordance with MIL-E-16400 and MIL-E-917 and shall maintain the cleaning solution temperature to within plus or minus 2°F from the set point. Means to set the temperature between 50 and 200°F and a temperature deviation indicator shall be readily accessible to the operator. The temperature sensor shall be of the Resistive Temperature Detector (RTD) type with a positive temperature coefficient, or a thermocouple.

3.4.5.5 Faucet and spray rinse. A swivel-type of faucet constructed for bulkhead mounting shall be provided. Hot and cold, potable water shall be piped to the faucet assemble by the installing facility.

3.4.5.5.1 Spray equipment. A pressurized spray rinse system shall be provided consisting of an airwater spray gun with flexible hoses located for use in the rinse tank. The pressurized rinse shall be sourced by hot water (130°F, 50 pounds per square inch (lb/in^2) operating pressure, 150 lb/in^2 test pressure and, compressed clean filtered air, 100 lb/in^2). The proportion of air and water shall be adjustable from all water and no air to no water and all air.

3.4.5.5.2 Plumbing connections. A stainless steel plate shall be provided that shall mount on the bulkhead. It shall support and accommodate the faucet's hot and cold water valves, the spray gun assembly and the air and water connections. The plumbing shall include bottom entry of hot and cold fresh water through a 1/2 inch brass tailpiece 12-inches long conforming to MIL-T-20168. The source of compressed air shall be through a l/4-inch copper tailpiece 12 inches long conforming to MIL-T-24107. Stainless steel drain valves that shall drain tanks in less than 5 minutes shall be provided. Valve operation shall be convenient to the operator. Tank and system drains shall be flexible connections employing high temperature chemical resistant hose for sound isolation.

3.4.5.6 Interconnecting piping. Interconnecting piping and fittings between pump, filter, heater and cleaning tank shall be of 316L stainless steel, seamless type (see 3.3.1). Valves not in contact with cleaning solution shall be of valve bronze. When in contact with cleaning solution, piping shall be 316L stainless steel, seamless type. When required, a 3-inch long (maximum) hose coupling may be utilized for connecting the tanks to the recirculating system for sound isolation purposes.

3.4.5.7 Strainer. A 316L stainless steel strainer accessible for servicing shall be installed in the system before the inlet of the pump to protect against large impurities.

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3.4.6 Consoles and cabinets. Unless otherwise specified (see 6.2.1), system components shall be housed in a simple cabinet or floor mounted console constructed of nonmagnetic stainless steel. Thermal and sound isolation shall be protected from water and cleaning solution. Hinged or lift off, stainless steel tank lids shall be provided. Doors or panels shall be provided for front access to allow servicing or replacement of system components. Modified console cabinet configurations shall be as specified (see 6.2.1).

3.4.7 Controls and indicators. Electrical controls, indicator lights, and meters shall be mounted flush on front of console, cabinet, or if provided, the control box. Electrical components shall be in accordance with MIL-C-2212 and MIL-E-917.

3.4.7.1 Controls. System components shall dictate the required controls and, if appropriate, shall include:

(a) Power “ON” switch
(b) Pump switch
(c) Heater switch
(d) Selector switch for manual or automatic mode (ultrasonic generator)
(e) Ultrasonic generator “ON” switch
(f) Timer setting (see 3.4.8)
(g) Temperature setting (see 3.4.5.4)

3.4.7.2 Indicators. System components shall dictate the required indicators and meters for the various functions with appropriate light levels or indicators that may include:

(a) Power available - white
(b) Sonics “ON” - green
(c) Pump “ON” - green
(d) Heater “ON” - red
(e) Sonics tank low liquid level - amber
(f) Clogged filter indicator - amber
(g) Sonic meter
(h) Temperature deviation meter (see 3.4.5.4.1)

3.4.8 Timer. A 30-minute timer mounted on the console or control box shall be required. The timer shall be self-resetting and allow the operator to select sonics “ON” time. At the end of the cycle with selector switch in the automatic mode, the timer shall turn off sonics and allow recirculating system including heater to start.

3.4.9 Fresh water fill piping. Fresh water fill piping to clean and rinse tanks shall be such that there shall not be a continuous water path between fill outlets and fluid in the tanks. This shall prevent possible contamination of the fresh water supply by bacteria or cleaning chemicals (see 3.4.5.5).

3.5 Electrical equipment. Unless otherwise specified (see 6.2.1), the system shall be wired for operation on power sources as follows:

(a) Sizes 1, 2, and 3 shall be wired for operation on 115 volt, single phase, 60 hertz (Hz), alternating current (at) power.

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(b) Size 4 shall be wired for operation on 440 volt, single or 3 phase, 60 Hz, ac power.
(c) Special sizes shall be wired as specified (see 6.2.1).

3.5.1 Wiring and circuit safety devices. Wiring and circuit safety devices shall be in accordance with MIL-C-2212 and MIL-E-917.

3.5.2 Recirculating systems. The recirculating system if required shall be automatically de-energized when sonic system is energized.

3.5.3 Fuses. The control circuit, pump motor, and heater shall be fused separately.

3.5.4 Magnetic controllers. Magnetic controllers shall be in accordance with MIL-C-2212 and shall have the following characteristics:

Type	- Across-the-line.
Performance	- Nonautomatic.
Protection	- Undervoltage protection.
Function	- Motor starting.

3.6 Training and installation. When specified (see 6.2.1), the services of a factory-trained representative shall be provided to supervise installation and train naval personnel at point of installation. This training service shall consist of necessary instructions in the proper operation and maintenance of the cleaning system.

3.7 Electromagnetic interference control. Unless otherwise specified (see 6.2.1), the equipment acquired under this specification shall conform to the requirements of MIL-STD-461.

3.8 Harmonic distortion. Unless otherwise specified (see 6.2.1), the equipment shall meet the requirements for harmonic distortion, in accordance with MIL-E-16400.

3.9 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate, permanently and legibly marked with the following Information, in accordance with MIL-STD-130, shall be securely attached to each enclosure.

(a) Manufacturer’s name
(b) Manufacturer’s model number
(c) Manufacturer’s serial number
(d) Contract number
(e) Date of manufacture
(f) National stock number (if applicable)

3.10 Reliability. The equipment shall have a minimum mean-time-between-failures (MTBF) of 5,000 hours (see 4.7). A failure is defined as any mal-function that either causes the loss of sonics, shuts off the unit, or causes water leaks.

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3.11 Maintainability. The equipment shall have a mean-time-to-repair of 2 hours.

3.12 Insulation resistance. The insulation resistance of the system shall be not less than 4 megohms (see 4.6.1).

3.13 Dielectric strength. Each individual enclosure shall withstand a dielectric strength test of 1900 volts alternating current (Vac) for not less than 7 minutes (see 4.6.2).

3.14 Workmanship. Workmanship of the equipment to be provided shall be in accordance with the requirements of this specification. Workmanship shall meet the applicable requirements of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in 4.5 through 4.9.

4.3.1 First article inspection report. When specified in the contract or order, a first article inspection report shall be prepared (see 6.2.2).

4.4 Quality conformance. Quality conformance shall consist of the examination and tests specified in 4.5 and 4.6.1 through 4.6.7 (except 4,6.4).

4.4.1 Sampling for quality conformance. Quality conformance tests shall be made on 100 percent of all production units.

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4.5 Examination. Each unit shall be inspected for conformance to the configuration and finish requirements of this specification.

4.6 Tests.

4.6.1 Insulation resistance. Before the application of test voltage, all circuits shall be thoroughly discharged by shorting them to the frame. The test voltage shall be applied between all electrically isolated circuits and between each circuit and ground. The measurements shall be made with an insulation-resistance indicating meter with a full scale reading of 100 or 200 megohms. An open circuit voltage of 500 volts direct current (Vdc) and a voltage of at least 450 Vac shall be applied across a resistance of 1 megohm. The temperature shall be measured and recorded. Insulation-resistance measurements shall be corrected to 77°F (25 degrees Celsius (°C)). Corrections shall be made on the basis of insulation resistance doubling for each 59°F (15°C) decrease in temperature. Corrections shall not be made for humidity. In the event, because its rated limits may be exceeded by the test voltage, components likely to be damaged by the insulation resistance test or dielectric test shall be disconnected from the circuit before testing. The system shall meet requirements specified in 3.12.

4.6.2 Dielectric strength. The complete system or individual enclosures and consoles of the system shall be tested in accordance with method 301 of MIL-STD-202 and shall meet the requirements specified in 3.13.

4.6.3 System reliability. Each ultrasonic cleaning system shall be continuously operated with the sonics "ON" for not less than 100 hours. If no interruptions or failures occur, the system shall be further tested as required in 4.6.5 and 4.6.7 for individual acceptance test on production units. Failure to complete 100 hours of continuous operation shall require starting the test over after correction of the malfunction.

4.6.4 Accelerated hydraulic filter cleaning. The accelerated hydraulic filter cleaning test shall be performed as follows:

(a) Filters: Three supramesh filter elements, Aircraft Porous Media, Inc., part no. AC0-2466E-62Q, or equivalent shall be contaminated with hydraulic oil, 2190 TEP, to the extent that each filter shall be loaded to 500 lb/in^2 differential pressure. The hydraulic oil shall be contaminated with the fuel contaminant specified in MIL-E-5007, except the grams of contaminants per 1000 gallons shall be reduced in direct proportion to the quantity of oil used.

(b) The following equipment and material shall be required:

(1) Distilled water in quantity sufficient to conduct the test.
(2) The cleaning agent shall be in accordance with type I of MIL-D-16791.
(3) Necessary equipment required to conduct dirt capacity and bubble point tests on the filters. Filter manufacturer's services may be required for these tests.

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(c) The cleaning solution in the ultrasonic cleaner under evaluation shall have a concentration recommended by the contractor and approved by the Government agency concerned and shall be heated to a temperature between 140 and 160°F. The solution shall be deaerated for 30 minutes before cleaning the filters, by energizing the sonic generator on the unit under evaluation.

(d) Three preloaded filters shall be cleaned one at a time in the cleaner under evaluation as specified in 4.6.4(a). The filters shall be cleaned for a duration of time not greater than 15 minutes. The three filters shall be cleaned one after another using the same solution. Solution shall be circulated following the cleaning operation of each filter.

(e) The filters shall be tested as specified in Field Service Report No. 10C, except cleaned dirt capacity for each filter shall be at least 70 percent of actual dirt capacity.

(f) The ultrasonic cleaner contractor shall conduct the above filter cleaning test in the presence of and to the satisfaction of the Government agency concerned.

4.6.5 Cleaning test, ceramic ring/graphite. Immediately following the hydraulic filter test (see 4.6.4), a cleaning performance test shall be performed. A detergent solution from the hydraulic filter test shall be used or a new solution shall be prepared to demonstrate the cleaning ability of the unit by cleaning six graphite coated ceramic rings as follows:

(a) Six each aluminum oxide, unglazed, porous surface rings 1-5/8 inches outside diameter (od) by 1-1/4 inches inside diameter (id) by l/8-inch thickness shall be used. One flat surface of each ring shall be contaminated by rubbing 2-gauge graphite lead against the ring's surface. The contaminated rings shall stand at room temperature for not less than 1 hour and not more than 4 hours before testing.

(b) Cleaning solution temperature shall be maintained at 150 ± 3°F.

(c) The solution shall be allowed to de-gas for 15 minutes.

(d) The rings shall be placed in a wire basket, with the contaminated side up and equally spaced.

(e) The basket with rings shall be immersed into the cleaning tank to a position 1 inch from the bottom. The sonics shall be activated for 1 minute.

(f) The basket with rings shall be removed from the tank, rinsed in clean water and observed for the amount of carbon removed by comparing the color of the previously contaminated surface to the color of the clean surface. Black graphite on the white ceramic surface shall cover less than 25 percent of the surface of each of the rings.

4.6.6 Loaded cleaning test. A load test shall be performed as follows: A dummy load of 1/2 to 1 inch stainless steel cubes shall be randomly (not oriented) distributed directly on the bottom of the ultrasonic tank. The load shall be at least 10 pounds of stainless steel per square foot of tank bottom. The ceramic ring cleaning test specified in 4.6.5 shall be repeated above the dummy load. At least 50 percent of the ring surface shall be free of the black graphite.

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4.6.7 Noise. A noise test of the cleaning system shall be conducted in accordance with the requirements of MIL-STD-740-1, grade B. The noise test shall be witnessed by a Government representative.

4.6.8 Loaded cleaning. A load test shall be performed as follows: A dummy load of 1/2 to 1 inch stainless steel cubes shall be randomly (not oriented) distributed directly on the bottom of the ultrasonic tank. The load shall be at least 10 pounds of stainless steel per square foot of tank bottom. The ceramic ring cleaning test specified in 4.6.5 shall be repeated above the dummy load. At least 50 percent of the ring surface shall be free of the black graphite.

4.6.9 Dry run. The ultrasonic tank shall be intentionally allowed to drain while the ultrasonic system is "ON". The unit shall run dry for 5 minutes without damage to the transducers or generator and upon refilling shall resume satisfactory operation.

4.6.10 Impact. This test shall be conducted with no liquid in the tank. A 2-pound steel ball shall be dropped five times from 18 inches above the tank bottom to randomly selected locations on the tank bottom. Cleaning efficiency shall not be impaired. As a result of this test, tank rupture and deformation shall be cause for rejection.

4.6.11 Electromagnetic interference control. The system shall be tested in accordance with MIL-STD-461.

4.6.12 Harmonic distortion. The system shall be tested in accordance with MIL-E-16400. Operating modes shall be tested.

4.6.13 Transducer bonding nondestructive. A nondestructive test shall be conducted to ensure that a minimum of 96 percent of the mating surface of the transducer stacks are metallically joined to the diaphragm plate.

4.6.14 Shock. The cleaning system shall be tested for conformance to the requirements specified in 3.3.10.

4.6.15 Vibration. The cleaning system shall be tested for conformance to the requirements specified in 3.3.11.

4.7 Reliability. Reliability assurance program in accordance with MIL-STD-781 and MIL-HDBK-217, shall be conducted to determine conformance to the requirements specified in 3.10.

4.8 Maintainability. In accordance with MIL-STD-470 and MIL-HDBK-472 maintainability evaluation of final system shall be performed before construction of production units to determine conformance to the requirements specified in 3.11.

4.9 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

5.1.1.1 Lumber and plywood. When specified (see 6.2.1), all lumber and plywood including laminated veneer material used in shipping container construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Level A and B	- Type II - weather resistant. Category I - general use.
Level C	- Type I - non-weather resistant. Category I - general use.

5.1.1.2 Fiberboard. When specified (see 6.2.1), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, and cleated fiberboard boxes shall meet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

5.1.2 Disassembly. When disassembly is required (see 3.2.3), disassembled or removed components or parts, except hardware which shall be reinstalled in the mating part to prevent hardware loss, shall be tagged, marked and the tags attached to each mating part. The tags and printing thereon shall be resistant to water, oil, and fading.

5.2 Preservation. Preservation shall be level A, or commercial as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. Cleaning and drying of all cleaning system surfaces, internally and externally, shall be by a process and procedure in accordance with MIL-P-116 which will ensure removal of all foreign material from the system and disassembled or removed components or parts.

5.2.1.2 Unit protection. Each system including disassembled or removed components or parts shall be unit protected to meet the requirements of MIL-P-116, method III (physical and mechanical protection). All filters or cartridges of the replacement or throwaway type shall be new. Disassembled or removed parts shall be unit packed in a folding, set-up or metal edged paperboard or fiberboard box, at the contractor's option meeting the unit container requirements of MIL-STD-2073-1, appendix F, Box closure shall be in accordance with the applicable box specification or appendix thereto, except that method V shall apply for class-weather resistant and method I using pressure sensitive tape for class-domestic or class-domestic/fire retardant fiberboard boxes. Each complete system including its disassembled or removed components or parts when applicable shall be individually unit packed within the shipping container selected by the contractor for the required level of packing (see 5.3).

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5.2.2 Commercial. Commercial packaging (cleaning, preservation, cushioning and unit pack) of the cleaning system, disassembled or removed components or parts when applicable, shall be in accordance with the requirements of ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2.1).

5.3.1 Levels A, B, and C containers. Cleaning systems preserved as specified (see 5.2) shall be packed in shipping containers, for the level of packing specified (see 6 .2.1), in accordance with MIL-STD-2073-1, appendix C, and herein. Unless otherwise specified (see 6.2.1), container selection shall be at the contractor's option.

5.3.1.1 Caseliners, closure, and gross weight.

5.3.1.1.1 Caseliners. Level A shipping containers shall be provided with waterproof caseliners in accordance with MIL-STD-2073-1.

5.3. 1.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto, except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with nonmetallic or tape banding, and domestic or fire retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.1.1.3 Weight. Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.2 Commercial. Cleaning systems preserved as specified (see 5.2), shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.2.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two, 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment and storage.

5.4 Marking.

5.4.1 Levels A, B, C, and commercial. In addition to any special marking required (see 6.2.1), interior (unit) packs and shipping containers shall be marked including bar coding in accordance with MIL-STD-2073-1, appendix F.

6. NOTES

6.1 Intended use. The ultrasonic cleaning systems covered by this specification are intended for use on Naval ships, in Naval shipyards and other military facilities. The systems shall be used for cleaning hydraulic filters, air filters, electronic and electromechanical assemblies, strainers, valves and other objects within their capacity.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required, if special, provide complete description (see 1.2 and 1.3).
(c) When first article inspection is required (see 3.1).
(d) If the cleaning system is for submarine application, provide hatch entry dimensions and size restrictions (see 3.2.3).
(e) If airborne noise requirements are different (see 3.2.4).
(f) If equipment is not provided with mounting holes and hardware compatible with sound isolation mounts (see 3.2.4.1).
(g) Painting, if other than specified (see 3,3.7).
(h) Cleaning tank power, dimensions and console (see 3.4.3 and 3.4.6).
(i) Rinse tank dimensions, if required (see 3.4.4).
(j) Recirculating filter and heater system, if required (see 3.4.5).
(k) Capacity and retention capability of cleaning system (see 3.4.5.2).
(l) Temperature controller, if required (see 3.4.5.4.1).
(m) Voltage requirements and special sizing (see 3.5).
(n) Training, if required (see 3.6).
(o) Electromagnetic interference control requirements, if other than specified (see 3.8 and 4.6.11).
(p) Harmonic distortion, if other than specified (see 3.8 and 4.6.12).
(q) Nameplate, if other than specified (see 3.9).
(r) When fire retardant requirements are required (see 5.1.1).
(s) Level of preservation and packing required (see 5.2 and 5.3).
(t) Container selection if other than contractor's option (see 5.3.1).
(u) Special marking required (see 5.4.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.3.1	First article inspection procedure	DI-T-4901	----
4.3.1	First article inspection report	DI-T-4902	----

(Data item descriptions related to this specification and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article, When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Sub-contracted material and parts. The preparation for delivery or packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment *is* shipped.

6.6 Technical manuals. The requirement for technical manuals should be considered when this specification is cited on a contract. If technical manuals are required, a contract exhibit must be prepared to fully describe statement of work criteria and delivery instructions, and cite the applicable technical manual specification. The technical manuals must be acquired by separate Contract Line Item Number (CLIN) in the contract.

6.7 Subject term (key word) listing.

Electrostatic precipitator cells
Magnetostrictive
Nickel laminate
Thermocouple
Transducers
Generator

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6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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(Project 4940-N541)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-24196A(SH)	CLEANING SYSTEM, ULTRASONIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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